

# Microwave assisted synthesis and characterization of end functionalized poly(propylene oxide) as model compounds

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Received 26 June 2007; received in revised form 29 October 2007; accepted 31 October 2007

Available online 6 November 2007

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## Abstract

End functionalized poly(propylene oxide)s with *n*-alkyl endgroups are synthesized by anionic ring opening polymerization of propylene oxide (PO) in bulk employing controlled microwave heating in sealed vessels. Different alcohols ranging from *n*-butanol to *n*-octadecanol are used as initiator and 2–10 propylene oxide units are added on average. A continuous decrease of internal pressure during the sealed vessel experiment reflects the consumption of PO monomer and the completion of the reaction is confirmed by a drop of the internal pressure to zero. The products are characterized by size exclusion chromatography (SEC), liquid chromatography at critical conditions (LCCC) and liquid adsorption chromatography (LAC). SEC with coupled density and refractive index (RI) detection provides information on the chemical composition along the molar mass distribution. LCCC allows a separation of the individual polymer homologous series according to their end groups. In LAC, the alkyl terminated chains elute later than the non-functional chains and can be separated to the baseline according to the number of repeat units.

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**Keywords:** End functionalized PPO; Microwave-assisted polymerizations; Anionic ring opening polymerizations; Model compounds for chromatography

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## 1. Introduction

Polypropylene glycols (PPG) are important in polymer chemistry since these have applications in polyurethanes synthesis [1,2], as hydraulic fluids, lubricants, surface active agents and chemical intermediates [3]. Despite their many applications described in the literature, not much attention is

given to end functionality of these polymers. Brisault [4] had reported the introduction of various end groups on commercially available diols. The end groups may influence their physical properties considerably. Moreover, monoalkyl ethers of PPG can serve as starting materials in the synthesis of ABC triblock systems, where A is the alkyl group, B the a polyoxypropylene chain, and C a hydrophilic block, such as polyoxyethylene. For this purpose, samples with the highest possible purity are desirable, i.e., the content of starting material (the alcohol) and chains with a different functionality

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(such as diols) must be as low as possible. To ensure the desired high purity of the products, suitable and reliable methods for their characterization must be available. For the development of such methods, a sufficient number of reference samples with different functionality and molar mass is required.

These samples should be prepared on demand, and their synthesis should be possible in a reasonably short time frame. Therefore, microwave-assisted polymerization is a feasible technique for this purpose [5–11]. Such reference materials may also be used for basic research on the chromatographic behaviour of polymers in general, which can serve to build up a database for predicting polymer chromatography [12].

In a recent paper we have described the factors affecting anionic ring opening polymerization of PO using PEG-MME as initiator [13] with controlled microwave heating in sealed vessels.

In this study we describe the synthesis of PPG monoalkyl ethers employing controlled microwave heating in sealed vessels and their characterization using different chromatographic techniques, such as size exclusion chromatography (SEC) with dual detection, liquid chromatography at critical conditions (LCCC) and liquid adsorption chromatography (LAC).

## 2. Characterization of the products

Functional polymers and oligomers can be characterized using different chromatographic techniques, which separate according to different criteria. The type of chromatographic separation is determined by the so-called interaction parameter  $c$ , which describes the interaction of a structural unit with the stationary phase [14].

For homopolymers without functional end groups, retention can be described by the interaction parameter  $c_A$  of the repeat unit A. In the case of functional polymers and block copolymers, which can be considered as equivalent [15], the interaction parameter  $c_B$  of the other block B (or the end group) has also to be taken into account. Typical examples are nonionic surfactants, such as fatty alcohol ethoxylates (FAE). The polymers to be analyzed in these investigations are also of the AB type (diblocks or monofunctionals). In this case, the repeat unit A is the oxypropylene (PO) unit and the alkyl end group can be considered as a chain with the methylene group as the repeat unit B. The influence of the other end group

(OH) is negligible in typical reversed phase systems. For ABA and BAB polymers (such as PEG diesters [15–17] and poloxamers [18,19]) the situation is more complicated. As these structures are not relevant in this study, they shall not be discussed here.

The adsorption interaction parameter  $c$  has been first introduced by de Gennes [20]. It has a meaning of the inverse correlation length of adsorption, which can serve as a characteristic of the structure of a macromolecule near the surface of an adsorbent. Positive values of the parameter  $c$  correspond to the regime of adsorption [21–23]. In liquid adsorption chromatography (LAC) it represents the average thickness  $H$  of a flat layer formed by a long macromolecule on a pore wall:  $H = c^{-1}$ .

Negative  $c$  values are characteristic of the situation, where adsorption interactions are small or absent: in size exclusion chromatography (SEC)  $c$  is negative.

At the critical adsorption point (CAP), when the entropy losses of a macromolecule in a pore are precisely compensated by an energy gain due to the adsorption, the interaction parameter equals zero, and the corresponding block becomes “chromatographically invisible” [24–27]. This is utilized in liquid chromatography under critical conditions (LCCC), which allows the separation of polymers according to their end groups or to other blocks [25,28–30]. It must be mentioned, that there is no sharp limit between these types of polymers: a longer alkyl end group can as well be considered as a short block [15].

In the case of block polymers the interaction parameters  $c_A$  and  $c_B$  of the individual structural units (A and B) may have different values. If both  $c_A$  and  $c_B$  are strongly negative, the molar mass distribution (MMD) can be determined by SEC, with retention decreasing with increasing molar mass, and only a minor influence of chemical composition. SEC yields the MMD of the entire molecule (provided that the calibration functions for A and B are comparable).

With dual detection [31–33] one may obtain information on the chemical composition along the MMD, as will be discussed later on.

SEC is, however, not capable of discriminating mixtures of homopolymers with different repeat unit or polymers with the same repeat unit, but different functionality.

At the CAP of the repeat unit A, the elution volume is independent of the chain length of the

block A. In this case, AB polymers can be separated according to the block B.

Basically, there are two options: if the interaction parameter  $c_B$  of the block B is negative, it will elute earlier than the homopolymer A in SEC order, if  $c_B$  is positive, the block copolymer elutes later than the homopolymer A in LAC order [18,19]. Obviously, the first situation will only be applicable, if the B block is sufficiently large. On the other hand, the second approach is limited to short blocks in the case of strong interaction (i.e. large values of  $c_B$ ).

Functional polymers can be separated by LCCC according to their end groups: all chains with the same end group elute as a narrow peak at the same volume, which will be the higher, the stronger the adsorption of the end group is.

There is, however, another possible situation: if the interaction parameter  $c_A$  of the repeat unit A is negative and the interaction parameter  $c_B$  of the end group B is positive, the individual oligomers elute in SEC order, but with elution volumes much larger than the void volume. This is utilized in liquid exclusion adsorption chromatography (LEAC), which allows an excellent separation of non-ionic surfactants under isocratic conditions [17,34–37].

If, however, both  $c_A$  and  $c_B$  are positive, a new situation may occur: at strong interaction of the end group and weak interaction of the repeat unit, different homologous series may be separated from each other, and at the same time, there may be a baseline resolution of the individual oligomers of each series. This situation is somewhat similar to LEAC, but with the opposite elution order.

### 3. Quantification in LC of AB copolymers

In the analysis of AB copolymers (whether random or block) the sensitivity of the detector for A and B may be very different. In this case, one will obtain erroneous results using just one detector.

With a combination of two concentration detectors (such as RI, UV and density detector [38]) with different response factors for the units A and B one may obtain important information on chemical composition and thus an accurate result.

As the polymers to be analyzed in this study do not have any UV-absorbing groups, the combination of density and RI detection is the method of choice. With amphiphilic copolymers, this works, however, only in SEC, where single mobile phases are used. In mixed mobile phases the preferential

solvation of the polymer molecules would require a third concentration detector, which is not available. The evaporative light scattering detector (ELSD) cannot be applied for this purpose because of its unclear response to copolymers.

As has been shown previously [31,39], response factors of density and RI detection depend on chemical composition and molar mass.

When a mass  $m_i$  of a copolymer containing the weight fractions  $w_A$  and  $w_B (=1 - w_A)$  of the components A and B elutes in the interval  $i$  of the peak, its composition ( $w_A$ ) can be calculated from the areas  $x_{i,j}$  obtained for this interval from both detectors, and the response factors  $f_{j,A}$  and  $f_{j,B}$ , wherein  $j$  denotes the individual detectors

$$\frac{1}{w_A} = 1 - \frac{\left(\frac{x_{i,1}}{x_{i,2}} * f_{2,A} - f_{1,A}\right)}{\left(\frac{x_{i,1}}{x_{i,2}} * f_{2,B} - f_{1,B}\right)} \quad (1)$$

From the composition of a fraction its mass can be calculated using Eq. (2)

$$m_i = \frac{x_{i,1}}{w_A * (f_{1,A} - f_{1,B}) + f_{1,B}} \quad (2)$$

This approach can be utilized, if the response factors  $f_{j,A}$  and  $f_{j,B}$  are sufficiently different [31–33,40–43].

In the analysis of functional polymers and oligomers another approach can be applied.

If the end groups are chemically different from the repeat unit, the response factors depend on molar mass:

$$f_i = f_\infty + \frac{K}{M_i} \quad (3)$$

wherein  $f_i$  is the response factor of a molecule with the molar mass  $M_i$ ,  $f_\infty$  the response factor of the repeat unit, and  $K$  is a constant describing the influence of the end group.

This allows an accurate quantification in SEC of functional oligomers with just one detector. The only limitation is, that Eq. (3) can only be applied, if the sign of the response factor does not change within the peak.

For functional polymers, both approaches are equivalent. In this study, we have preferred the dual detector approach (using Eqs. (1) and (2)).

LCCC is typically performed in mixed mobile phases, hence preferential solvation has to be taken into account. This is not the case with the ELSD, but this detector is highly problematic in the analysis of lower oligomers, which may be not detected at

all [44]. On the other hand, it is the only detector, which can be used to detect small amounts of homopolymers, as these elute in LCCC together with the solvent peak.

If the individual oligomers can be sufficiently separated by isocratic LAC and identified, a quantitative determination is possible: under isocratic conditions, density or RI detection can be applied, and for these detectors, the response factors can be calculated for each oligomer using Eq. (3).

## 4. Experimental

### 4.1. Polymerizations

Propylene oxide (PO) and all alcohols (butanol, hexanol, octanol, decanol, dodecanol, tetradecanol, hexadecanol and octadecanol) were purchased from Fluka (Buchs, Switzerland). Sodium hydride (60% dispersion in mineral oil) from Aldrich was used as received. Lower alcohols (butanol–decanol) were dried over anhydrous sodium sulfate while higher alcohols were used as received.

The polymerization of PO was performed by anionic ring opening using alcohols and sodium hydride as initiator. All polymerizations were performed in 2–5 ml (filling volume) microwave process vials (Biotage AB), which were oven dried under vacuum prior to use. The alcohol and NaH were added to the vials which were subsequently sealed

with an aluminum crimp top and Teflon septum. The calculated volume of PO was added with a syringe through the septum. The quantity of sodium hydride was 0.1% of total product, the quantities of alcohols and PO in different experiments depended upon intended average number (2–10) of PO units in the chain (calculated by molar ratios). The sealed microwave process vials were subsequently introduced into the cavity of the single-mode microwave reactor (Biotage Initiator Eight EXP 2.0, absorbance level: high).

At reaction temperatures of 160 °C the initial pressure in the microwave process vial was typically in a range between 5 and 20 bar and decreased during the polymerization as PO was consumed. Due to the fact that PO has a boiling point of only 34.3 °C the autogenic pressure generated by this starting material in combination with the online pressure monitoring of the microwave reactor [45] did provide a very convenient method for monitoring the conversion in these solvent-free polymerization processes.

All microwave-assisted polymerizations were carried out at 160 °C to complete conversion, as indicated by a drop of the internal pressure from initially 5 to 20 bar to zero. Typically reactions take 50–80 min for complete conversion (see Fig. 1) while the same experiment performed under conventional heating takes almost 5 h. After polymerization the active end was neutralized by adding an equimolar quantity of acetic acid.

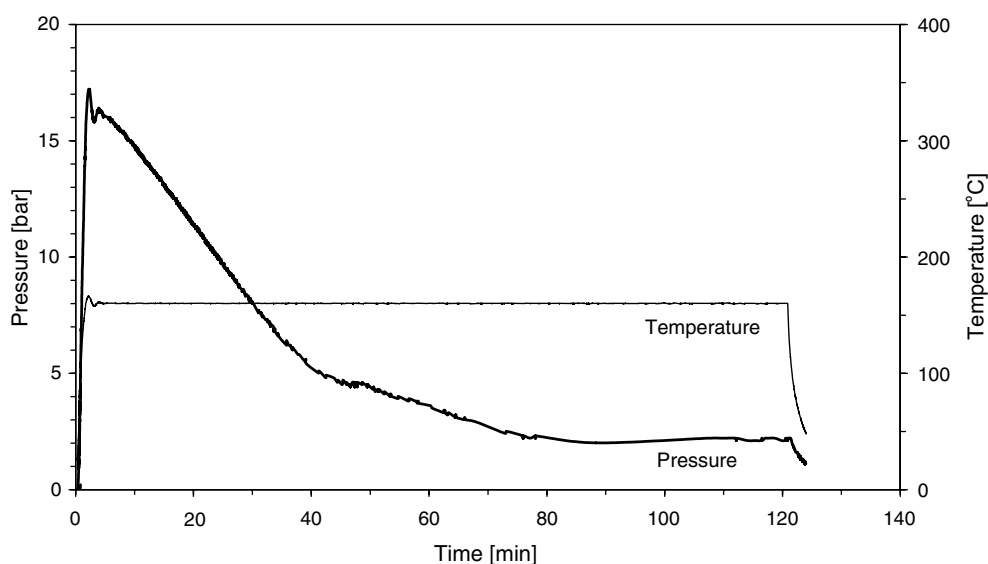


Fig. 1. Internal pressure and temperature during the polymerization under microwave irradiation.

## 4.2. Chromatography

These investigations were performed using a density detector (according to the mechanical oscillator principle [46,47]) DDS70 (CHROMTECH, Graz, Austria) in all chromatographic systems. Data acquisition and processing was performed using the software CHROMA, which has been developed for the DDS70.

In SEC, we used a modular system consisting of a Gynkotek 300 C pump, a VICI injector equipped with a 100  $\mu$ l sample loop, two column selection valves Rheodyne 7060 (Rheodyne, Cotati, CA, USA), a density detection system DDS 70 (Chromtech, Graz, Austria) coupled with a Bischoff 8110 refractive index detector (Bischoff, Leonberg, Germany). SEC measurements were performed on a 7.8  $\times$  600 mm Phenogel 100A column (Phenomenex, Torrance, CA, USA) or a 7.8  $\times$  300 mm PLgel MIXED E column (Polymer Laboratories, Church Stretton, Shropshire, UK) at a flow-rate of 1.00 ml/min. Sample concentrations were 3.0–10.0 g/l. The columns were calibrated with PPGs from Sigma–Aldrich and FLUKA (Buchs, Switzerland). Molar mass distributions were calculated using CHROMA.

In LCCC, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA) at a flow rate of 0.5 ml/min. Samples were injected using an autosampler Spark SPH 125 Fix (Spark Holland, Emmen, The Netherlands) equipped with a 20  $\mu$ l loop. A refractive index detector RI 2414 (Waters) or a SEDEX 45 ELSA (Sedere, France) was connected to the DDS 70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. Evaporator temperature: 30 °C. All separations in LCCC were performed on a Jordi Gel DVB 500 RP column (100% poly-divinylbenzene; 250  $\times$  4.6 mm; particle diameter = 5  $\mu$ m; pore size = 500, from Jordi, Bellingham, MA, USA).

In LAC the mobile phase was delivered by a JASCO 880 PU pump (from Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 0.5 ml min<sup>-1</sup>. Samples were injected manually using a Rheodyne 7125 injection valve (from Rheodyne, Cotati, CA, USA) equipped with an 50  $\mu$ l loop. A Waters 2414 refractive index detector (Waters, Milford, MA, USA) was connected to the DDS 70. Columns were connected to two column selection valves (Rheodyne 7060, from Rheodyne, Cotati, CA, USA). All LAC separations were performed on a Prodigy ODS3 column (from Phenomenex, Torrance, CA, USA): sil-

ica-based octadecyl phase; 250  $\times$  4.6 mm; particle diameter = 5  $\mu$ m; pore size = 100, surface area: 450 m<sup>2</sup>/g, carbon load: 15.5%.

In all systems, the columns and density cells were placed in a thermostatted box, in which a temperature of 25.0 °C was maintained for all measurements in LCCC and LAC and 30.0 °C in SEC.

Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

The solvents (chloroform, THF, acetone and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany) and Merck (Darmstadt, Germany). PEG and PPG standards were purchased from Sigma–Aldrich and FLUKA (Buchs, Switzerland).

## 5. Results and discussion

In the case of end-functionalized PPG one may expect (besides residual alcohol) mainly two other homologous series as side products: PPG diols (resulting from chain transfer to traces of water in the starting materials) and allylic PPG (which is formed by chain transfer to the monomer) [13,48].

The residual alcohol can be identified and determined quantitatively by SEC on a column with narrow pores, as it is sufficiently smaller than the first oligomer.

Some unreacted alcohol can be found in samples with a low ratio of PO to alcohol, while it completely disappears in samples with (on average) 5 or more PO units, as can be seen in Fig. 2.

SEC with dual detection can be applied to determine the PO content along the MMD: in a low molecular sample with a short alkyl end group, where the individual oligomers can be at least partially resolved, a stepwise increase of the PO content (corresponding to the chemical composition of the oligomers) can be observed, as is shown in Fig. 3.

Higher molecular samples or those with a longer alkyl end group (in which the individual oligomers cannot be resolved any more) show also the increase of PO with molar mass, and at the low molecular end one may see the presence or absence of residual alcohol as is shown in Fig. 4.

Fig. 5 shows the MMD of two different hexadecyl-PPGs with a different number of repeat units (5 and 10 on average). As can be seen, both samples contain only traces of the starting alcohol.

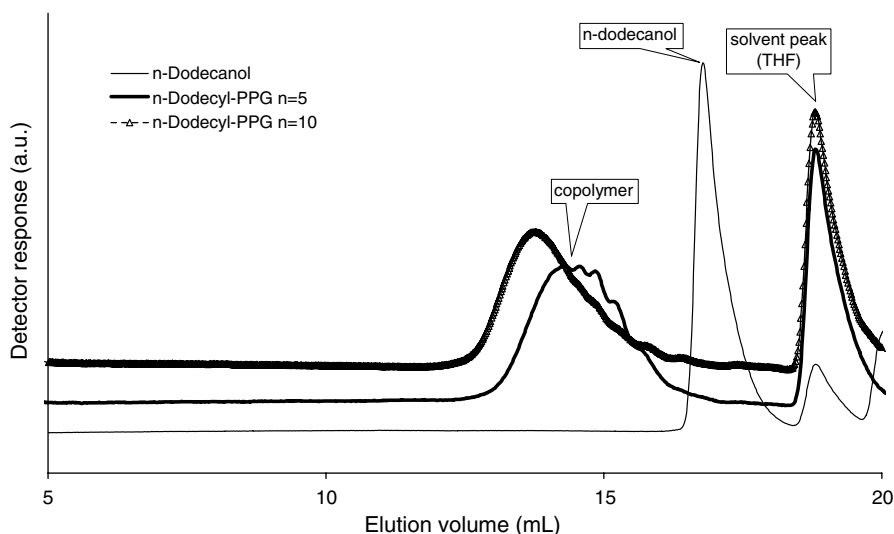


Fig. 2. Overlay of SEC chromatograms with *n*-dodecanol as starting block (density detector).

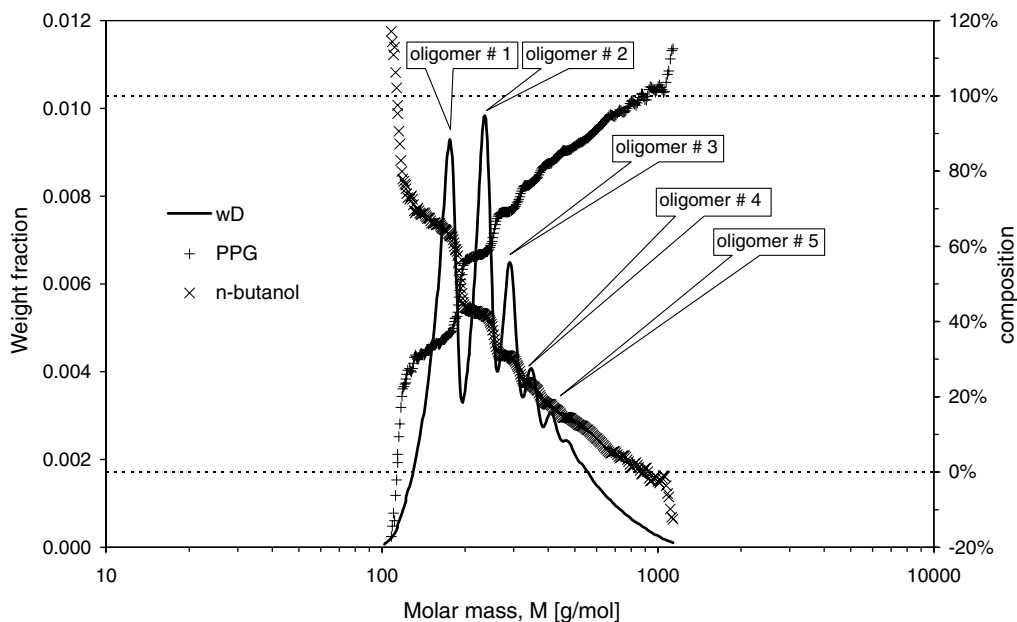


Fig. 3. MMD and chemical composition of a product with *n*-butanol as starting block, as obtained from SEC with dual detection.

Of course, PPG diols and allylic PPG will not be identified, and if they would be present in considerable amounts, they would seriously affect the accuracy of the MMD obtained from SEC.

This information can only be obtained from LCCC. At the CAP for PO, the elution volume of each series is independent of molar mass. PPG diols and allylic PPG are separated from each other (due to the adsorption of the allyl group) [13].

If the alkyl group is long enough, it is adsorbed more strongly than the allylic group, and the alkyl-terminated PPG can also be separated from the allylic PPG.

This is a different situation from the EO-PO diblocks described recently [13,18,19], where the EO block elutes in the exclusion mode (earlier than the diols).

Fig. 6 shows the elution volumes of the individual homologous series with different molar mass



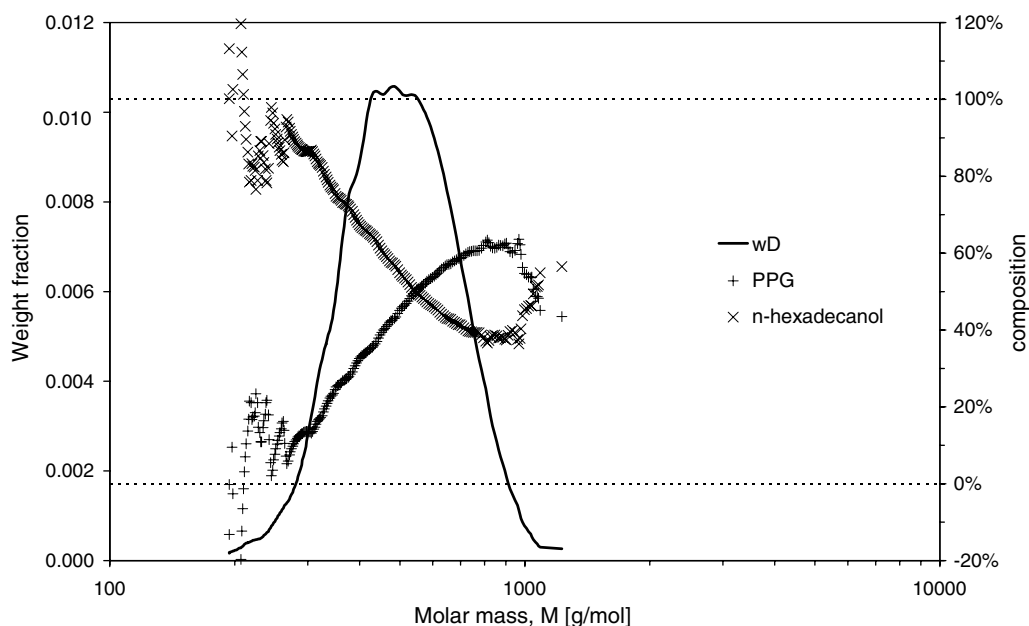


Fig. 4. MMD and chemical composition of a product with *n*-hexadecanol as starting block, as obtained from SEC with dual detection.

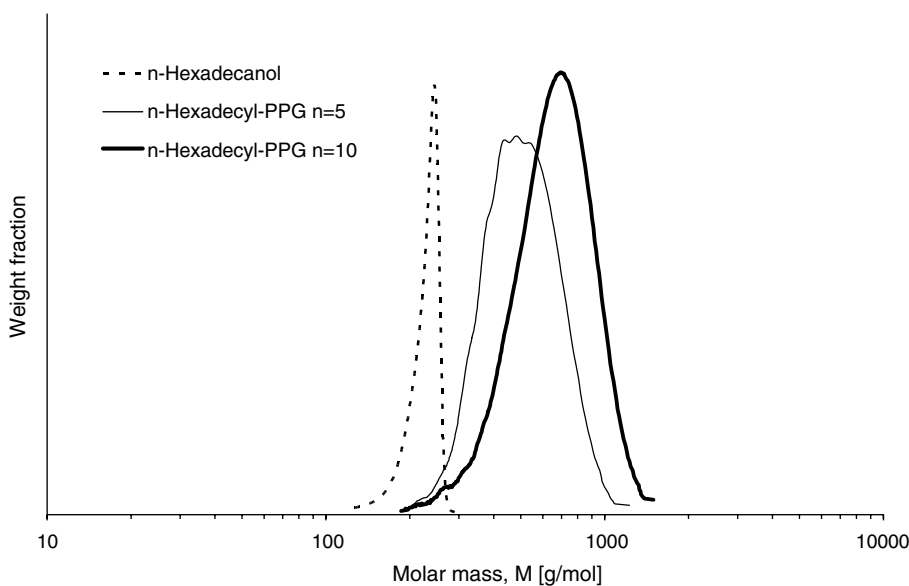


Fig. 5. Overlay of MMD of PPGs with hexadecyl end group, as obtained from SEC with dual detection.

( $M$  vs.  $V_e$ ). As can be seen, the elution volumes of PPGs with the same end groups are almost independent of molar mass (except for those with C16 and C18 end groups), which indicates, that this mobile phase composition does not exactly correspond to the critical point, but it is close enough to the CAP to allow a good separation of PPGs with an end group of 10 or more carbon atoms from the

allyl-terminated PPGs and the PPG diols. The peaks of polymers with short alkyl end groups (like butyl and hexyl) may overlap with the peak of the allyl-PPGs, but PPGs with longer alkyl end groups (such as decyl, dodecyl, hexadecyl and octadecyl) elute later and are thus clearly separated from the allyl-terminated PPGs. All alkyl PPGs are, however, clearly separated from the diols.

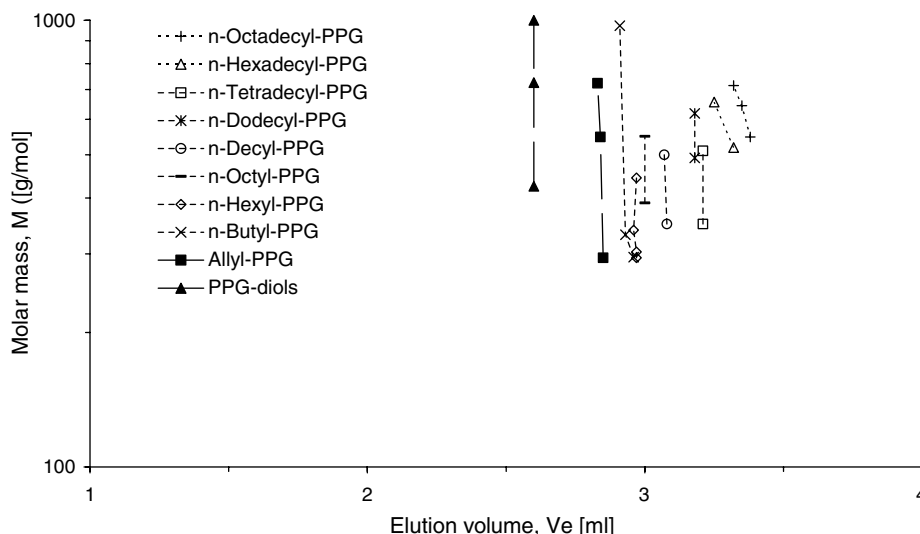


Fig. 6. LCCC: Molar mass vs. elution volume, as obtained on the Jordi column in 80 wt% THF-water, flowrate 0.5 ml/min, detection: ELSD.

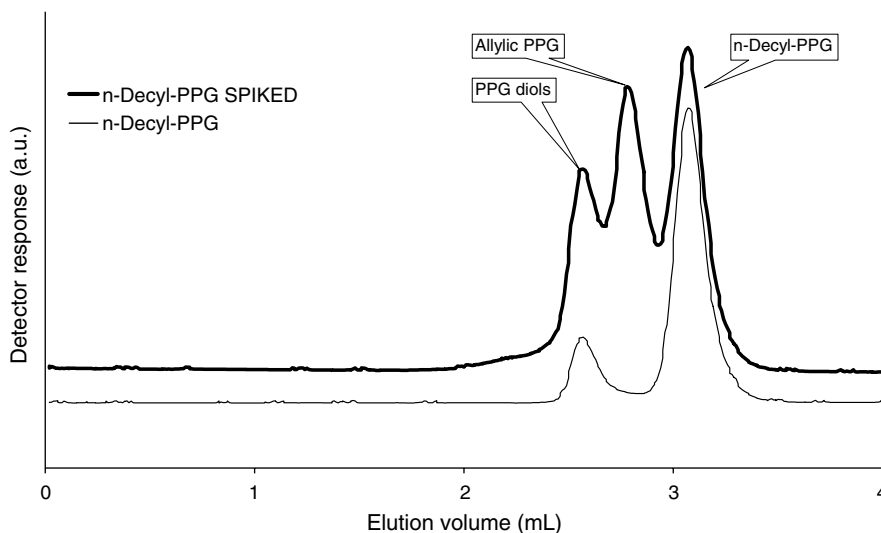


Fig. 7. LCCC of products with decyl end group, as obtained on the Jordi column in 80 wt% THF-water, flowrate 0.5 ml/min, detection: ELSD.

Fig. 7 shows chromatograms of a sample with decyl end groups, which obviously contains diols, but no allylic PPG. This sample was spiked with allylic PPG, which had been prepared by propoxylation of allyl alcohol (in the same way as the alkyl PPGs). As can be seen, allyl-terminated PPG appears as a sufficiently resolved peak.

Fig. 8 shows a comparison of two samples with octadecyl end groups: while the sample with  $n = 10$  PO units contains considerable amounts of

the other homologous series, the one with  $n = 5$  contains only traces of them.

While the diols can be minimized by drying reagents and glassware carefully, the allylic chains cannot be easily avoided, as they are formed by a chain transfer reaction to the monomer. As has been shown previously [13,48], this reaction can be more or less suppressed by a sufficiently high concentration of hydroxy groups. In the case of the alcohols this is fulfilled quite well: the amount of



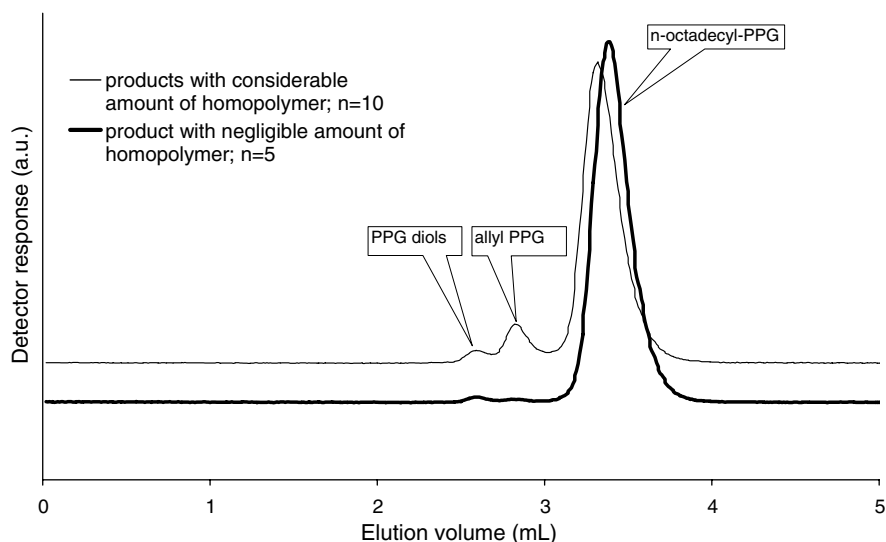


Fig. 8. LCCC of products with octadecyl end group, as obtained on the Jordi column in 80 wt% THF-water, flowrate 0.5 ml/min, detection: ELSD.

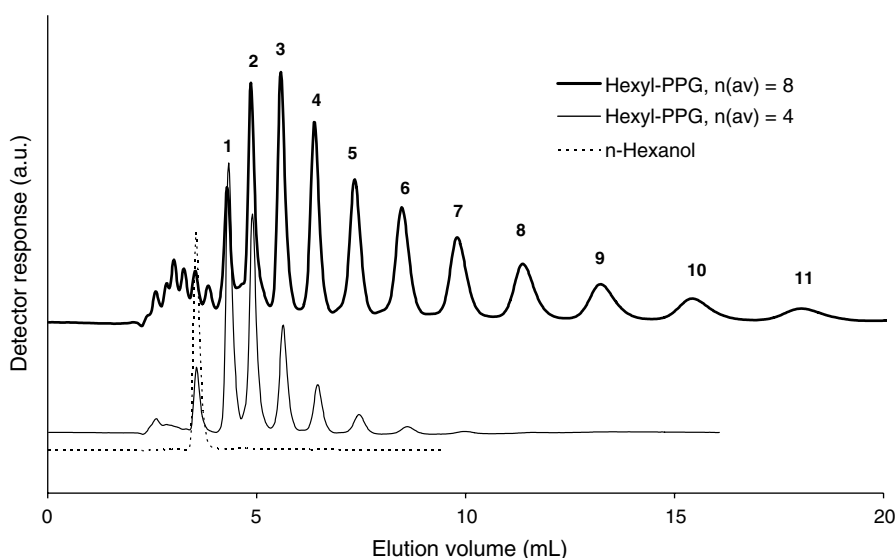


Fig. 9. Overlay of LAC chromatograms of mono-hexyl PPGs and hexanol. Prodigy ODS3, 70.83 wt% acetone–water, RI detection.

allyl-terminated PPG is quite low in most of the samples.

The main problem in these separations is quantification: if LCCC is run in THF-water, the response factors of density and RI detection for PPG (regardless their functionality) are poor, as the difference in density and refractive index between sample and mobile phase is too small. Hence the ELSD has to be used, which is much more sensitive. The lowest

oligomers of PPG and lower alcohols (up to decanol) are, however, not detected by ELSD.

Additional information can be obtained by using a mobile phase, in which both interaction parameters (PO and alkyl) are positive. This is the case on ODS columns in acetone–water mobile phases with about 70–85 wt% acetone. Depending on the nature of the end groups, different mobile phase compositions must be applied. If the interaction of

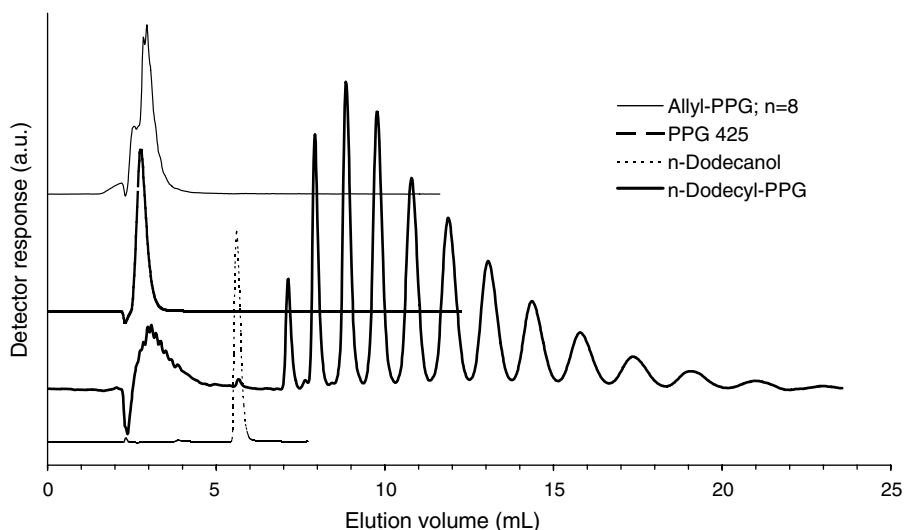


Fig. 10. Overlay of LAC chromatograms of dodecyl PPG, allyl PPG, PPG diol, and dodecanol. Prodigy ODS3, 81.22 wt% acetone–water, RI detection.

the end group is strong, the interaction of the repeat unit may be rather weak. For such functional polymers, a baseline separation of the individual oligomers in LAC mode is achieved, where this would not be possible with the homopolymers.

In Fig. 9 the chromatograms of two hexyl-terminated PPGs and hexanol are shown. As can be seen, the hexyl PPGs elute in LAC order [49] with a baseline separation of the individual oligomers. All peaks of the C6 series can be identified (as indicated by the numbers above each peak). The first series of peaks (which contains the allyl PPGs) is separated from the hexyl-PPGs (except for hexanol). In this mobile phase, RI detection can be applied, which makes quantification much easier [50].

Fig. 10 shows a comparison of chromatograms obtained on the Prodigy column in 80.22 wt% acetone: for mono-dodecyl PPG a baseline resolution of all oligomers is achieved, while mono-allyl PPG and PPG diols elute close to the solvent peak and do not overlap with the dodecanol peak. Consequently, all oligomers of dodecyl-PPG can be quantified from RI detection, as will be shown in another paper.

Moreover, these chromatograms yield the interaction parameters of the PO unit may be determined in the range of rather weak interaction, which could not be achieved with non-functional chains. The data thus obtained can be used to build up a data base for predicting polymer chromatography, which will be the subject of a forthcoming publication.

## 6. Conclusion

End functionalized (alkyl) PPO with an average number of 2–10 PO units and good purity can be synthesized in bulk using controlled microwave dielectric heating in sealed vessels. The required total reaction times at 160 °C are only 50–80 min.. This makes it very feasible and very convenient to produce model compounds for LAC. These products are sufficiently pure to be used as starting diblock for synthesis of very interesting amphiphilic triblock architectures. A combination of SEC with coupled density and RI detection with LCCC is a very useful tool for complete characterization of end functionalized PPG.

## Acknowledgements

M.I.M. thanks the Higher Education Commission of Pakistan for a Ph.D. scholarship. Mag. Jennifer M. Kremsner is acknowledged for some experimental assistance. We also thank Biotage AB (Uppsala, Sweden) for providing the microwave instrumentation.

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